

mudah.my

property listings

Group: Fast&Furious

GOE JIE YING A23CS0224

NAWWARAH AUNI BINTI NAZRUDIN A23CS0143

YASMIN BATRISYIA BINTI ZAHIRUDDIN A23CS0201





Project *Background*

- Popular online marketplaces in Malaysia covering vehicles, electronics, jobs, and properties
- This project focuses on the property listings category
- Collecting this data manually is not practical due to the large volume spread across hundreds of pages
- A web crawler was built to automatically collect the data
- The collected data was used to compare the performance of Pandas, Polars, and DuckDB

Objectives



1. To build a web crawler to collect property listing data from Mudah.my across all states in Malaysia



2. To collect at least 100,000 records with 16 details per listing



3. To clean and organize the collected data for performance benchmarking and analysis



4. To measure processing time, CPU usage, memory usage, and throughput for Pandas, Polars, and DuckDB



5. To compare the results and identify which tool performs the best for large-scale data processing



Target Website Overview

Search page

Mudah.my > Property for Sale > Properties for Sale in Malaysia

BuyRentApartment Directory

Malaysia

Search Cities like Subang jaya

SEARCH

All Properties

Price Range (RM)

Tenure Type

Land Title

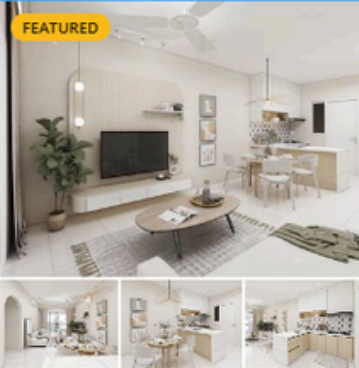
Sale Type

More filters

216,535 results found: All Properties For Sale in Malaysia

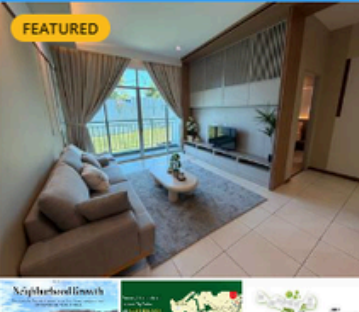
All (216,535)By Owner (16,074)By Agents (200,461)

FEATURED



Condominium For Sale, listed 15 Apr
RM 250,000
Setapak, Kuala Lumpur
850 sq.ft 3 Bed 2 Bath Freehold
Setapak Freehold Condo Monthly 1100 !!! Cash Back 40K
CONTACT SELLERby Property agent

FEATURED



New Apartment For Sale, listed 11 hours ago
RM 286,000
E Residence @ Telipok, Telipok
1,029 sq.ft 3 Bed 2 Bath Leasehold
RM1300+/Month, Affordable New 3-Bedroom Apartment @ Telipok, Zero Depo

SAVE SEARCH

Popular Locations in Malaysia

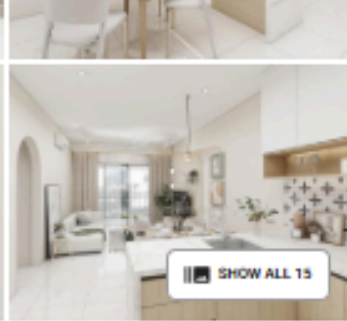
SelangorJohorPenangShow More

Popular Property Type in Malaysia

ApartmentsHousesCommercial PropertiesShow More

Popular Properties by

Listing Details Page



View SimilarSHOW ALL 15

Condominium For Sale, posted 15 Apr
RM 250,000
Setapak Freehold Condo Monthly 1100 !!! Cash Back 40K, Setapak, Kuala Lumpur
850 sq.ft 3 Bed 2 Bath

Seller Highlights by Mudah AI

New Condition

Low Deposit

Seller Offers

2 Car Parks

Near MRT/LRT

Nearby Essentials

Easy Highway Access

Swimming Pool

Show More

This highlight is AI-generated based on the seller's listing details

Description

Setapak Freehold Condo Monthly 1100 !!! Cash Back 40K
Setapak New Launch Promotion
Location: Setapak, Kuala Lumpur
Freehold Condo | Walking Distance to MRT & LRT
https://setapakrm1400newproject.wasap.my
Unit Types
2 - 4 Bedrooms
650 - 1000 sqft
SHOW MORE

Property Details

Property Type	Carpark	Floor Range
Condominium	2	Medium
Tenure Type	Land Title	
Freehold	Non Bumi Lot	

Facilities

Parking

Swimming Pool

Barbeque area

Club house

Security

Playground

Minimart

Jogging Track

Lift

Gymnasium

Multipurpose hall

Contact Agent

WhatsApp

Chat

012362****

Adrine Ang

Property agent
U PLUS REALTY
Firm: E - 11565, REN 73720
All ads from this advertiser (3 For Sale)

Mortgage Estimation

RM 996
Estimated monthly installment (Interest rate 4.0%)

Share This Ad

Facebook

WhatsApp

Email

Twitter

Telegram

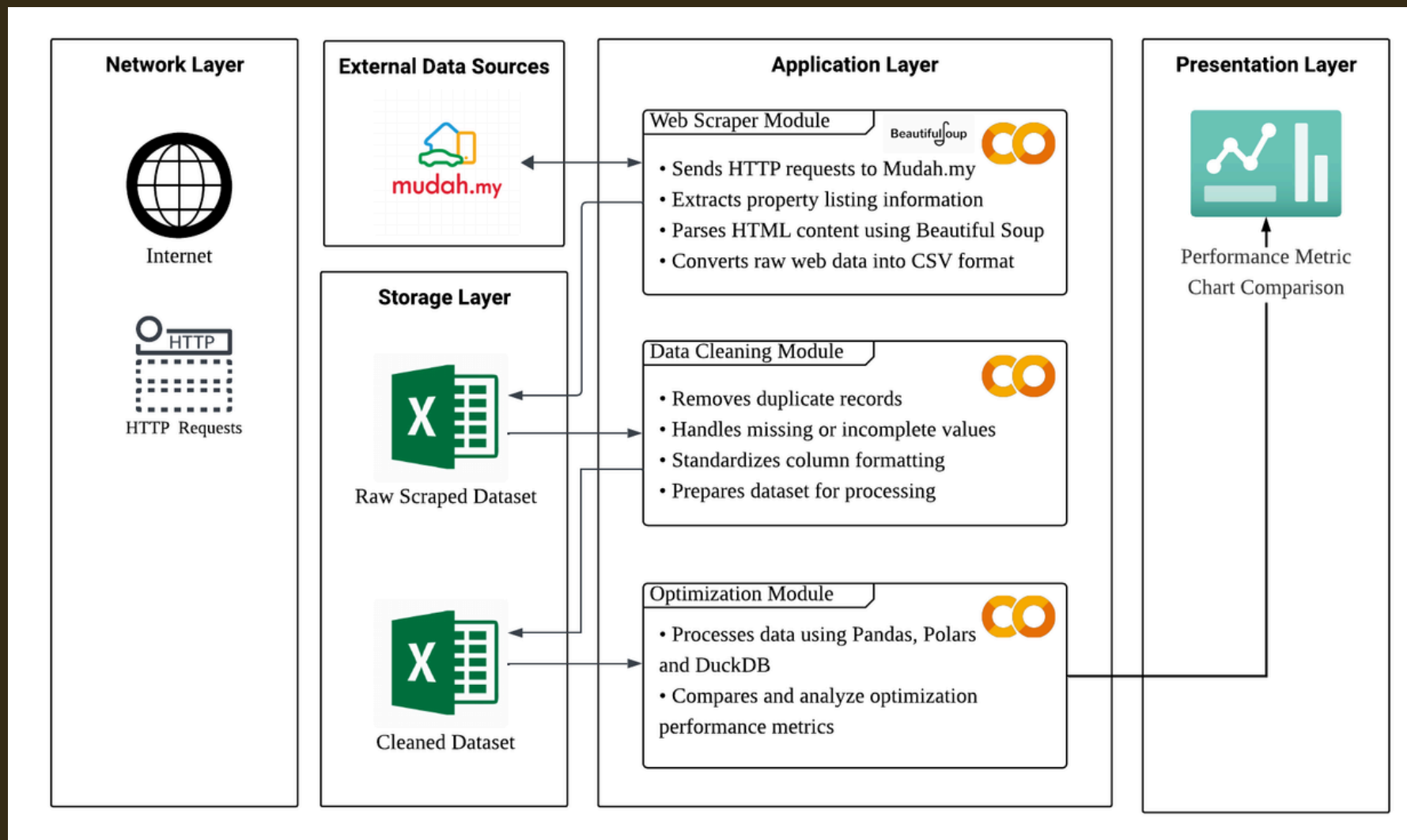
Line

Copy

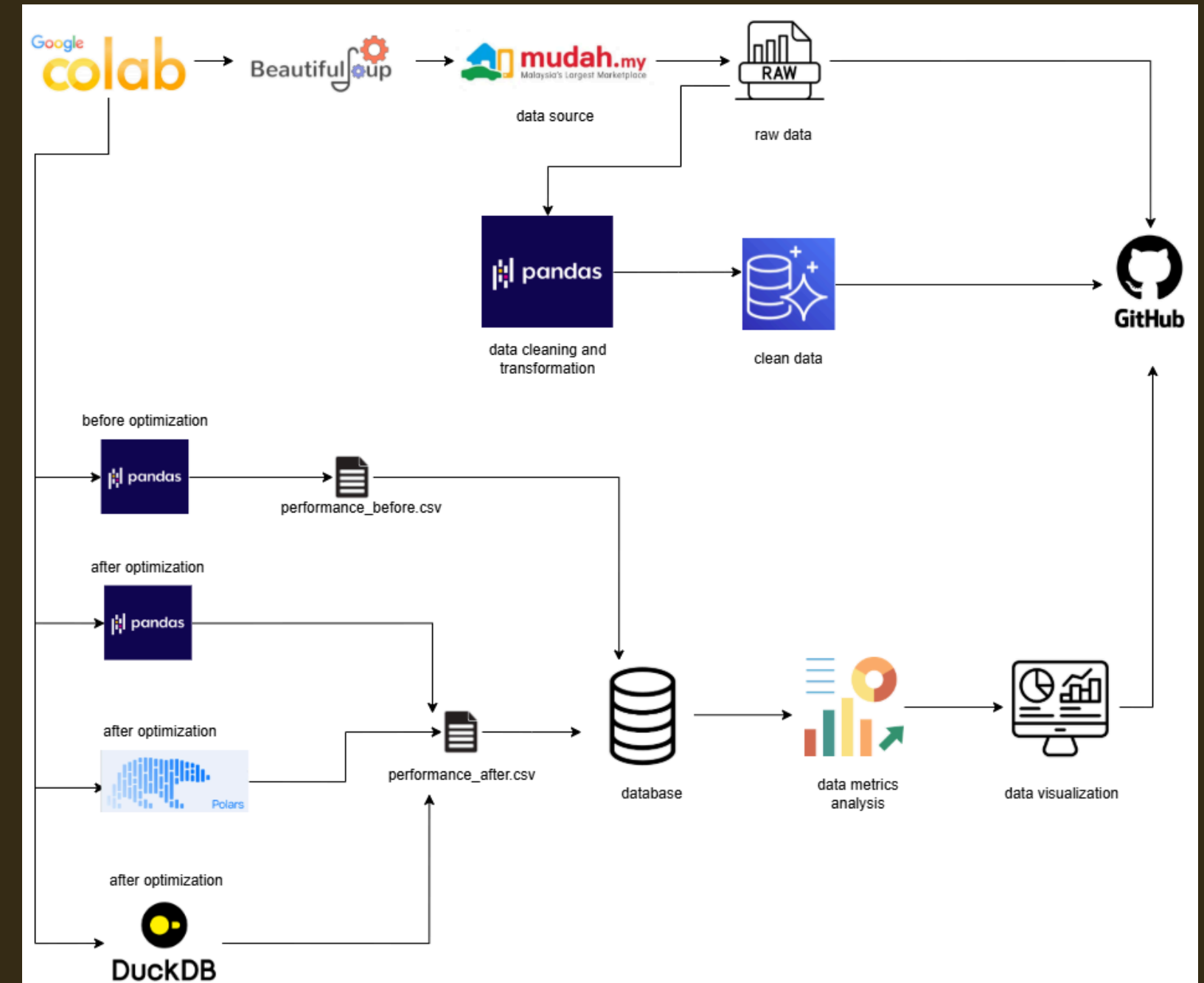
Is this ad suspicious or inappropriate?
Report this Ad (Lia-ID: 113997175)

System Design & Architecture

System Architecture

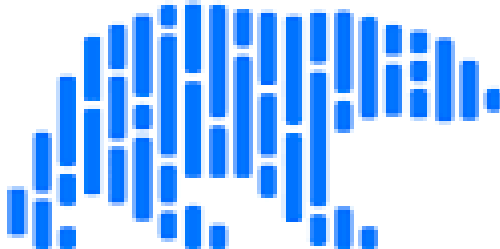


Tools and Frameworks Used Architecture

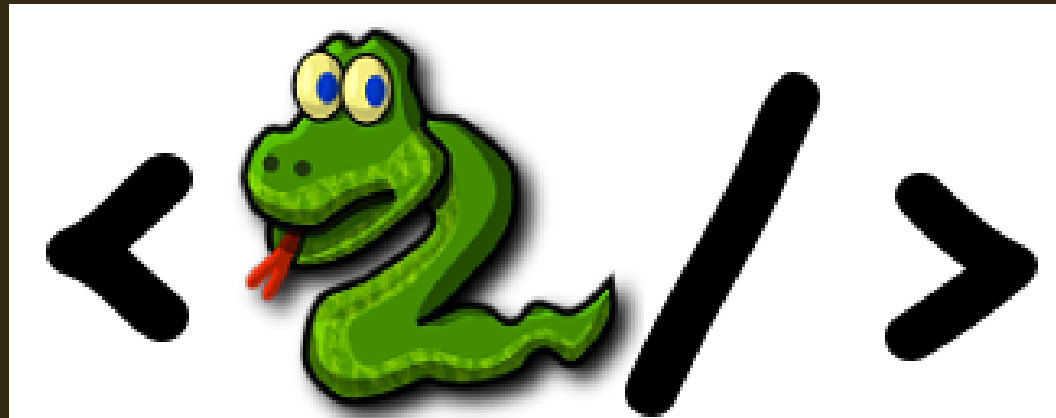


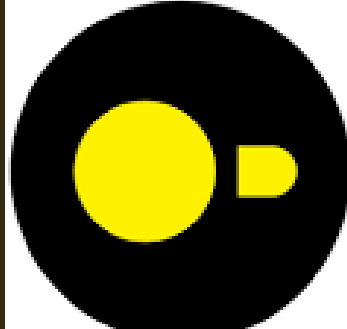
Tools & Frameworks Used

BeautifulSoup

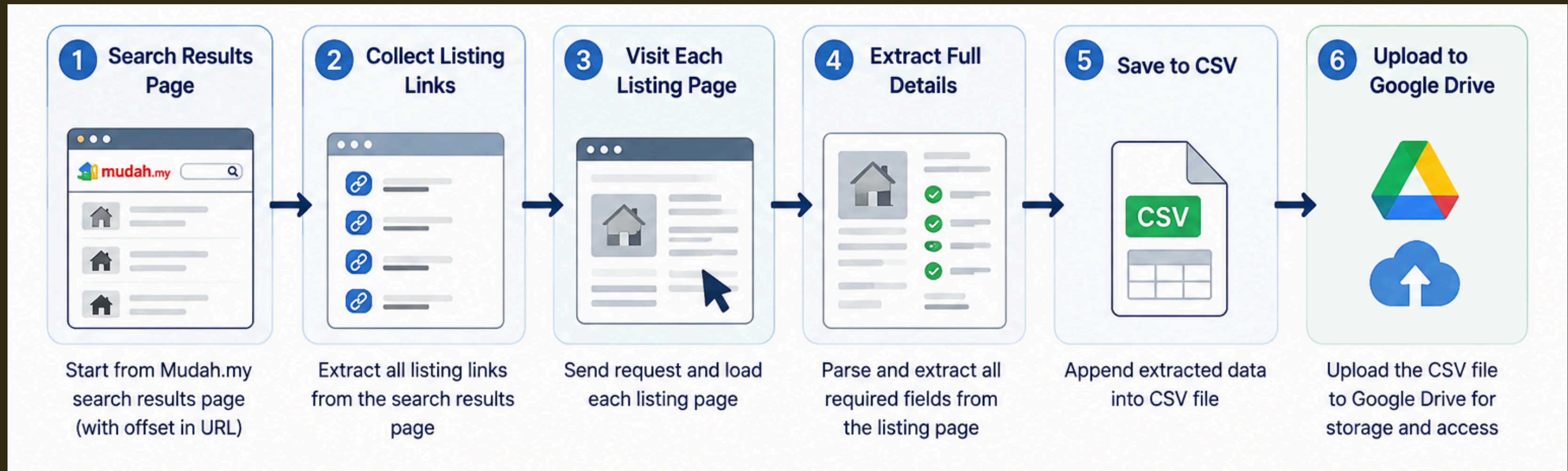
 Polars

 pandas



 DuckDB

Crawling methodology



Raw Dataset Structure



Collected from
Mudah.my across
all states in Malaysia



Combined into
a single file with
16 columns before
cleaning

No.	Column	Description
1	listing_id	Unique ID for each listing
2	property_type	Type of property
3	description	Description written by seller
4	location	Area of the property
5	state	State in Malaysia
6	price_rm	Price in Ringgit Malaysia
7	mortgage_est_rm	Estimated monthly mortgage
8	land_title	Land title type
9	size_sqft	Size in square feet
10	bedrooms	Number of bedrooms
11	bathrooms	Number of bathrooms
12	tenure	Freehold or leasehold
13	seller_type	Agent or individual seller
14	seller_name	Name of the seller
15	url	Link to the listing page
16	img_url	Link to the listing image

Raw Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	listing_id	property_type	description	location	state	price_rm	mortgage_	land_title	size_sqft	bedrooms	bathrooms	tenure	seller_type	seller_name	url	img_url		
2	1.15E+08	New Service	200	KLCC	Kuala Lumpur	700000	2789	Non Bumi	700	2	2	Leasehold	Agent		https://www	https://cdn.rnudah.com/images/		
3	1.12E+08	Apartment	[â€¦] Book	Teratai Me	Kuala Lumpur	185000	737	Non Bumi	650	3	1	Freehold	Agent		https://www	https://cdn.rnudah.com/images/		
4	1.14E+08	New Apartment	KL Sungai	PSV 1 Resi	Kuala Lumpur	410000	1634	Malay Resi	1376	4	3	Leasehold	Agent		https://www	https://cdn.rnudah.com/images/		
5	1.1E+08	Zero-Lot B	[FULLY	Taman Me	Kuala Lumpur	3100000	12353	Non Bumi	5270	5	6	Leasehold	Agent		https://www	https://cdn.rnudah.com/images/		
6	1.14E+08	1-storey Te	END LOT	Sri Damans	Kuala Lumpur	750000	2989	Non Bumi	1400	4	3	Freehold	Agent		https://www	https://cdn.rnudah.com/images/		
7	1.15E+08	Apartment	Pangsapuri	Pangsapuri	Kuala Lumpur	210000	837	Bumi Lot	650	3	1	Leasehold	Agent		https://www	https://cdn.rnudah.com/images/		
8	1.15E+08	Condominium	Series	Kuchai Lar	Kuala Lumpur	330000	1315	Non Bumi	680	2	2	Freehold	Agent		https://www	https://cdn.rnudah.com/images/		
9	1.15E+08	Service Res	Dual Key	Arte Cheras	Kuala Lumpur	300000	1195	Non Bumi	650	2	2	Freehold	Agent		https://www	https://cdn.rnudah.com/images/		
10	1.14E+08	2-storey Te	FREEHOLD	Taman Me	Kuala Lumpur	589000	2347	Non Bumi	840	3	2	Freehold	Agent		https://www	https://cdn.rnudah.com/images/		
11	1.15E+08	2-storey Te	2 Storey	Wangsa M	Kuala Lumpur	950000	3786	Non Bumi	1850	4	3	Freehold	Agent		https://www	https://cdn.rnudah.com/images/		
12	1.14E+08	2-storey Te	FACING	Pandan Peta	Kuala Lumpur	695000	2770	Non Bumi	1400	4	3	Leasehold	Agent		https://www	https://cdn.rnudah.com/images/		
13	1.15E+08	Apartment	Shop	Taman Bukit	Kuala Lumpur	170000	677	Bumi Lot	710	3	1	Leasehold	Unknown		https://www	https://cdn.rnudah.com/images/		
14	1.14E+08	Condominium	TERMURA	Suasana L	Kuala Lumpur	420000	1674	Non Bumi	1142	4	2	Leasehold	Agent		https://www	https://cdn.rnudah.com/images/		
15	1.15E+08	Condominium	Condomi	Sentul Utara	Kuala Lumpur	289000	1152	Bumi Lot	874	3	2	Leasehold	Agent		https://www	https://cdn.rnudah.com/images/		
16	1.15E+08	Residential	6,275 sqft	Kampung E	Kuala Lumpur	3250000	12951	Non Bumi Lot				Freehold	Agent		https://www	https://cdn.rnudah.com/images/		
17	1.15E+08	Flatforsale	Rumah	Sri Kota, B	Kuala Lumpur	190000	757	Bumi Lot	550	2	1	Freehold	Unknown		https://www	https://cdn.rnudah.com/images/		
18	1.13E+08	Flatforsale	Kos	Flat Panda	Kuala Lumpur	130000	518	Bumi Lot	650	2	1	Leasehold	Unknown		https://www	https://cdn.rnudah.com/images/		
19	1.1E+08	Apartment	Rumah	Desa Satu	Kuala Lumpur	145000	578	Non Bumi	650	3	2	Freehold	Unknown		https://www	https://cdn.rnudah.com/images/		

Data cleaning

```
...  — Fixing AUNI —
      Shape      : (31393, 16)
      Dtypes OK : listing_id=object, price_rm=int64, bedrooms=float64, bathrooms=float64

      — Fixing GOE —
      Shape      : (39348, 16)
      Dtypes OK : listing_id=object, price_rm=float64, bedrooms=float64, bathrooms=float64

      — Fixing YASMIN —
      Shape      : (30678, 16)
      Dtypes OK : listing_id=object, price_rm=int64, bedrooms=float64, bathrooms=float64

      — Fixing RENT —
      Shape      : (10201, 16)
      Dtypes OK : listing_id=object, price_rm=int64, bedrooms=float64, bathrooms=float64

=====
      Combined shape : (111620, 16)
      Dtypes:
listing_id      object
property_type   object
description      object
location        object
state           object
price_rm        float64
mortgage_est_rm float64
land_title      object
size_sqft       float64
```

raw_data

```
=====
      CLEANING SUMMARY REPORT
=====
Raw rows      : 111620
Cleaned rows   : 100442
Rows removed   : 11178

Columns       : ['listing_id', 'property_type', 'description', 'location', 'state', 'price_rm', 'mortgage_est_rm', 'land_title', 'size_sqft']

Remaining nulls:
listing_id      0
property_type   0
description      0
location        0
state           0
price_rm        0
mortgage_est_rm 0
land_title      0
size_sqft       0
bedrooms        0
bathrooms       0
tenure          0
seller_type     0
seller_name     0
url             0
img_url        0
dtype: int64
```

cleaned_data

Cleaned Dataset

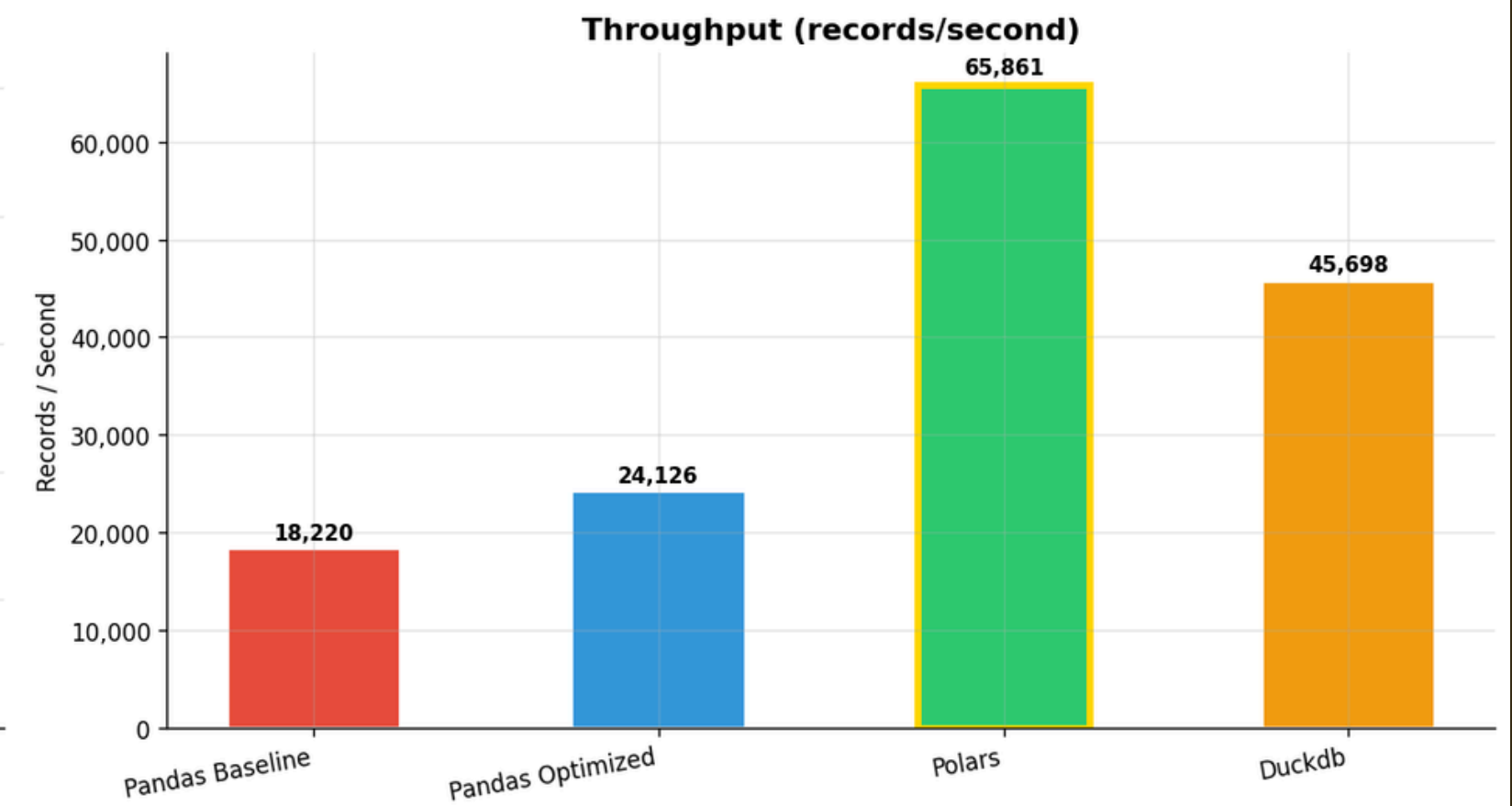
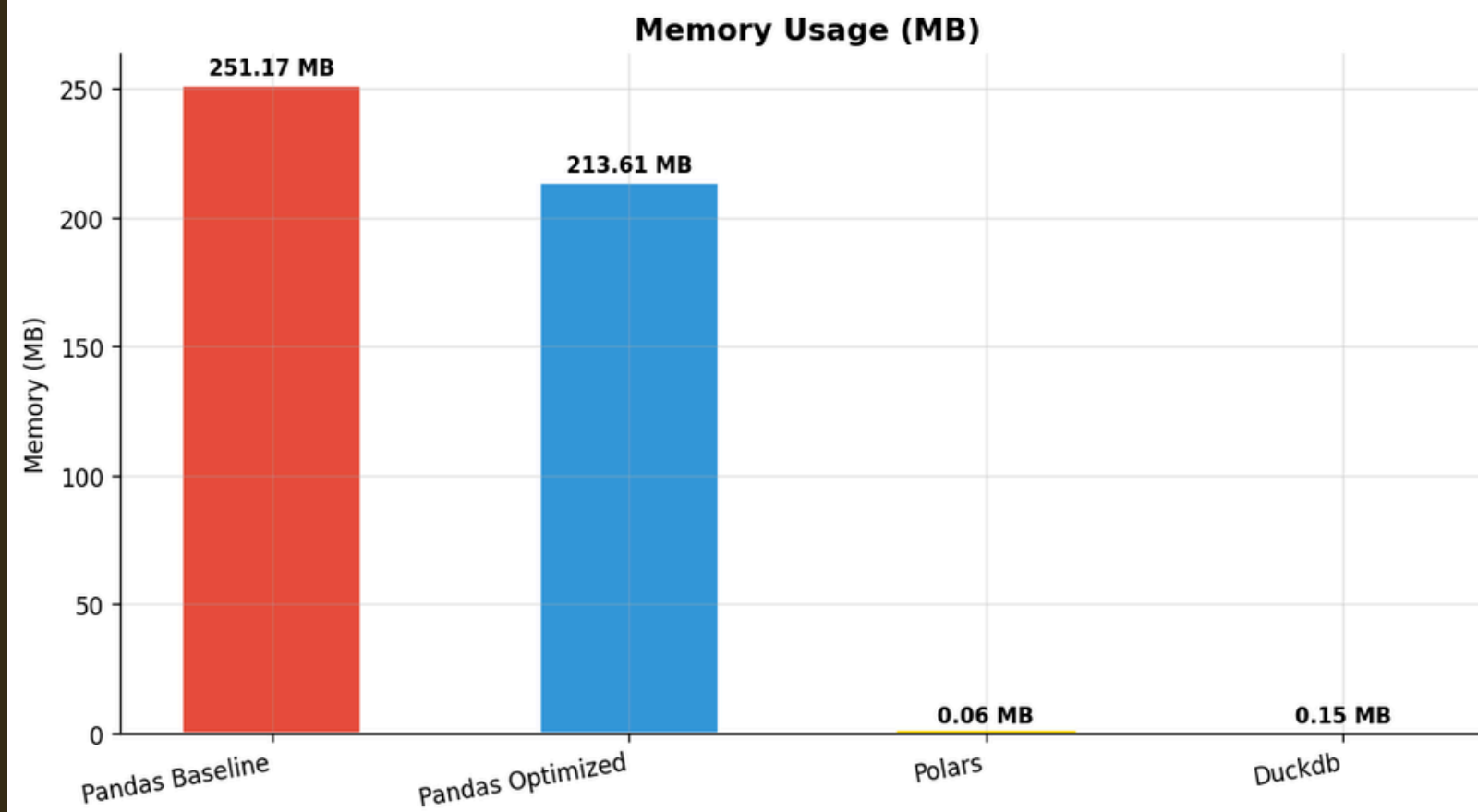
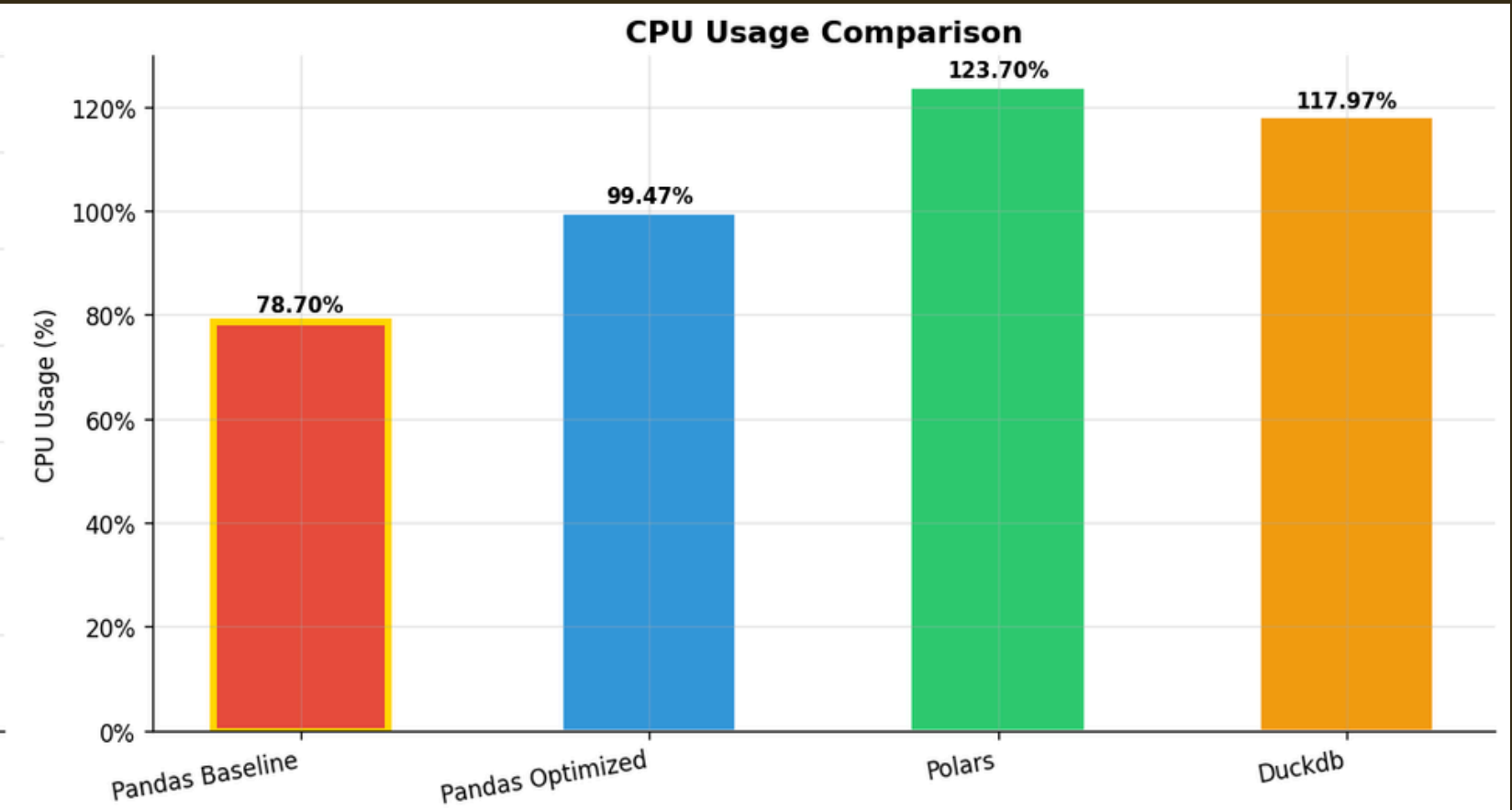
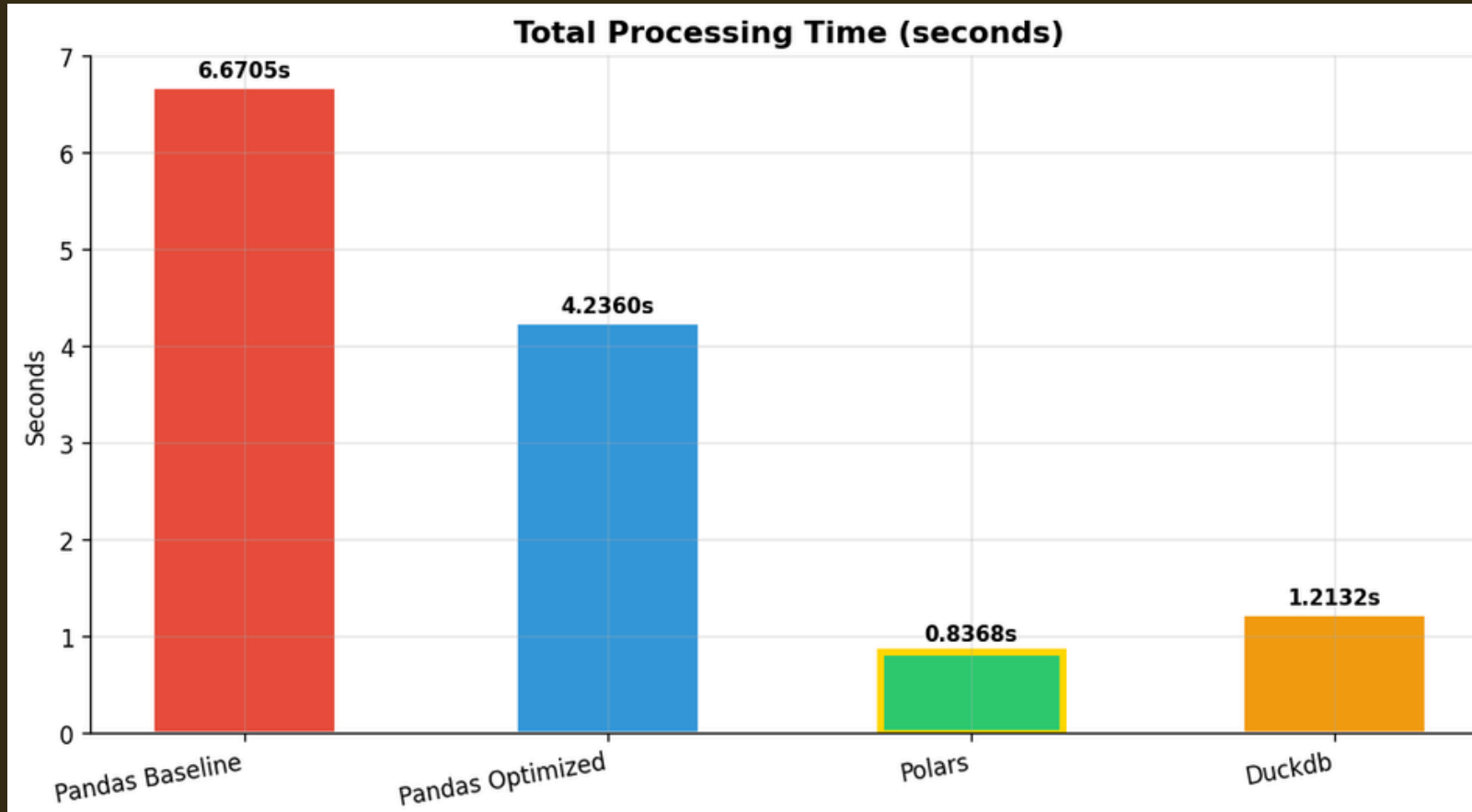
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
1	listing_id	property_type	description	location	state	price_rm	mortgage	land_title	size_sqft	bedrooms	bathrooms	tenure	seller_type	seller_name	url	img_url				
2	1.15E+08	New Service	200 meter	KLCC	Kuala Lumpur	700000	2789	Non Bumi Lot	700	2	2	Leasehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/479a15				
3	1.12E+08	Apartment	[â€œ] , Booki	Teratai Me	Kuala Lumpur	185000	737	Non Bumi Lot	650	3	1	Freehold	Agent	N/A	https://www.rnudah.com/property/1.12E+08	https://cdn.rnudah.com/images/plain/351e22				
4	1.14E+08	New Apartment	KL Sungai I	PSV 1 Resi	Kuala Lumpur	410000	1634	Malay Resi	1376	4	3	Leasehold	Agent	N/A	https://www.rnudah.com/property/1.14E+08	https://cdn.rnudah.com/images/plain/bc635f				
5	1.1E+08	Zero-Lot B	[FULLY FU	Taman Me	Kuala Lumpur	3100000	12353	Non Bumi Lot	5270	5	6	Leasehold	Agent	N/A	https://www.rnudah.com/property/1.1E+08	https://cdn.rnudah.com/images/plain/9c3b1f				
6	1.14E+08	1-storey Te	END LOT R	Sri Damans	Kuala Lumpur	750000	2989	Non Bumi Lot	1400	4	3	Freehold	Agent	N/A	https://www.rnudah.com/property/1.14E+08	https://cdn.rnudah.com/images/plain/e92246				
7	1.15E+08	Apartment	Pangsapuri	Pangsapuri	Kuala Lumpur	210000	837	Bumi Lot	650	3	1	Leasehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/fdc89a				
8	1.15E+08	Condominium	Series Cor	Kuchai Lar	Kuala Lumpur	330000	1315	Non Bumi Lot	680	2	2	Freehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/de6091				
9	1.15E+08	Service Res	Dual Key	Arte Cheras	Kuala Lumpur	300000	1195	Non Bumi Lot	650	2	2	Freehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/db68f1				
10	1.14E+08	2-storey Te	FREEHOLD	Taman Me	Kuala Lumpur	589000	2347	Non Bumi Lot	840	3	2	Freehold	Agent	N/A	https://www.rnudah.com/property/1.14E+08	https://cdn.rnudah.com/images/plain/336055				
11	1.15E+08	2-storey Te	2 Storey Te	Wangsa M	Kuala Lumpur	950000	3786	Non Bumi Lot	1850	4	3	Freehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/b7d5f3				
12	1.14E+08	2-storey Te	FACING O	Pandan P	Kuala Lumpur	695000	2770	Non Bumi Lot	1400	4	3	Leasehold	Agent	N/A	https://www.rnudah.com/property/1.14E+08	https://cdn.rnudah.com/images/plain/8ff2bc				
13	1.15E+08	Apartment	Shop apart	Taman Bukit	Kuala Lumpur	170000	677	Bumi Lot	710	3	1	Leasehold	Unknown	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/34d787				
14	1.14E+08	Condominium	TERMURAI	Suasana L	Kuala Lumpur	420000	1674	Non Bumi Lot	1142	4	2	Leasehold	Agent	N/A	https://www.rnudah.com/property/1.14E+08	https://cdn.rnudah.com/images/plain/b1a2e8				
15	1.15E+08	Condominium	Condominium	Sentul Utara	Kuala Lumpur	289000	1152	Bumi Lot	874	3	2	Leasehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/08b1fd				
16	1.15E+08	Residential	6,275 sqft	Kampung E	Kuala Lumpur	3250000	12951	Non Bumi Lot	1400	3	2	Freehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/9191d7				
17	1.15E+08	Flatforsale	Rumah ðŸ	'Sri Kota, B	Kuala Lumpur	190000	757	Bumi Lot	550	2	1	Freehold	Unknown	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/fed0bb				
18	1.13E+08	Flatforsale	Kos rental	Flat Panda	Kuala Lumpur	130000	518	Bumi Lot	650	2	1	Leasehold	Unknown	N/A	https://www.rnudah.com/property/1.13E+08	https://cdn.rnudah.com/images/plain/e97b7c				
19	1.1E+08	Apartment	Rumah flat	Desa Satu	Kuala Lumpur	145000	578	Non Bumi Lot	650	3	2	Freehold	Unknown	N/A	https://www.rnudah.com/property/1.1E+08	https://cdn.rnudah.com/images/plain/4d5394				
20	1.14E+08	Service Res	[Fully Furn	Sentul Poir	Kuala Lumpur	430000	1714	Bumi Lot	850	3	2	Freehold	Agent	N/A	https://www.rnudah.com/property/1.14E+08	https://cdn.rnudah.com/images/plain/301404				
21	1.15E+08	Service Res	865 sf Avail	Parc 3, Ch	Kuala Lumpur	680000	2710	Non Bumi Lot	865	3	2	Leasehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/261fb4				
22	1.15E+08	Service Res	990 sf Avail	South View	Kuala Lumpur	720000	2869	Non Bumi Lot	990	3	2	Freehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/3cf94f				
23	1.15E+08	Flatforsale	Indah Cour	Damai Cour	Kuala Lumpur	199000	793	Non Bumi Lot	729	3	1	Freehold	Agent	N/A	https://www.rnudah.com/property/1.15E+08	https://cdn.rnudah.com/images/plain/45f8f6				
24	1.13E+08	Commercial	KL Sentral	KL Sentral	Kuala Lumpur	2300000	9165	Non Bumi Lot	1400	3	2	Freehold	Agent	N/A	https://www.rnudah.com/property/1.13E+08	https://cdn.rnudah.com/images/plain/83c774				

Comparison

This table presents the comparison of key performance metrics, including total processing time, CPU usage, memory usage and throughput, for both the baseline and optimized methods across three benchmark runs.

method	run	time_sec	cpu_%	memory_mb	throughput	avg_time_sec	avg_cpu_%	avg_memory_mb	avg_throughput	speedup_vs_baseline
pandas_baseline	1	11.1974s	47.0%	251.16	8,970					
	2	4.1934s	91.7%	251.16	23,952	6.6705s	78.7%	251.17	18,220	1.00x
	3	4.6205s	97.4%	251.17	21,738					
pandas_optimized	1	3.7298s	100.0%	213.61	26,930					
	2	5.0461s	98.4%	213.61	19,905	4.2360s	99.5%	213.61	24,126	1.57x
	3	3.9322s	100.0%	213.61	25,544					
polars	1	0.9320s	113.7%	0.09	58,641					
	2	0.8334s	124.7%	0.04	65,582	0.8368s	123.7%	0.06	65,861	7.97x
	3	0.7450s	132.7%	0.04	73,360					
duckdb	1	1.3341s	107.2%	0.15	40,965					
	2	1.0186s	136.4%	0.15	53,658	1.2132s	118.0%	0.15	45,698	5.50x
	3	1.2869s	110.3%	0.15	42,470					

Visualization



Winner 



Polaris

Challenge and Limitation

Website restrictions

Initial websites blocked scraping (anti-bot & Cloudflare)



Limited data access

Only first 250 pages per category could be accessed despite large number of listings.



Data collection redesign

Crawling was split by state and category (sale/rent) to reach required dataset size.



Insufficient initial dataset

First dataset (6 columns) was not suitable for benchmarking, so re-crawling was required (final: 16 columns).



Data quality issues

Missing values in fields like bedrooms, bathrooms, and seller_name due to incomplete listings.



Time constraints

Revisions and re-crawling caused delays in optimization and documentation stages.



Future Work

- **Incorporating Real-Time crawling**

Implement a scheduled or incremental crawling pipeline that periodically updates the dataset to track price changes, new listings and removed listings over time.

- **Multimodal data integration**

As the current dataset consists mostly of structured text and numeric fields. Future work could incorporate unstructured data such as property listing images to build a multimodal pipeline.

- **Benchmarking additional processing engines**

Extend the benchmark tools by including other modern data processing engines such as Apache Arrow, or Vaex so that could provide a more comprehensive comparison across a wider range of tools and hardware configurations.

Conclusion

Key Results

All optimized tools outperformed
Pandas baseline
Polars 🏆 best overall — 0.84s,
~0.06MB memory, ~65,861 records/sec
DuckDB — strong in memory efficiency
& query stability

Conclusion

- Polars = best for large CSV data (lazy evaluation, multi-threading, low memory)
- DuckDB = best for SQL lovers (queries raw CSV without loading into memory)
- Tool choice significantly impacts performance on large-scale datasets



thank
you

